Student number: Solutions

Semester 2, 2024

Mathematics and Statistics

MAST20030_24_SM2 - Differential Equations

Reading time: 15 minutes

Writing time: 3 hours

Exam Venue Rules

- Do not communicate with other students in the exam venue
- During writing time, write your student number on the cover sheet, and on any script books or answer pages
- Blue scrap paper is for informal working notes only and will be collected and destroyed at the end of the session
- No part of the exam paper may be removed from the venue
- Ensure you only have permitted materials on your desk. The permitted materials are listed on this cover sheet
- Raise your hand if you require assistance
- Turn off your phone prior to commencement of reading time. Phones must be fully powered off

Standard Permitted Materials

- Identification card (Hardcopy student ID, Passport or Australian Government-issued photo ID)
- Pencils, pens, erasers
- Clear or unlabelled bottle of water

Additional Permitted Materials

- No other permitted materials

Exam Instructions

- During reading time, students may not write, type, or select answers
- Students must use a blue or black ball-point pen
- Students may use a pencil for diagrams
- Show all working or calculation steps for each question
- Write your answers in the designated answer boxes provided for each question
- If you require additional space for your answers, use the pages at the back of the exam paper
- Marks and weightings differ per question. Ensure you verify the marks for each question
- Next to each question you will find its assigned marks/weight.

Special Exam Instructions

None

Exam Weighting

None

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Question 1 (24 marks)

Consider the system

$$\begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix} = \boldsymbol{A} \cdot \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} \qquad \text{where} \qquad \boldsymbol{A} = \begin{bmatrix} -\alpha & 1 \\ 1 & -\alpha \end{bmatrix}, \quad \alpha \in \mathbb{R}.$$

(a) Find the eigenvalues of $\boldsymbol{A}$.

$$\begin{vmatrix} -\alpha-\lambda & 1 \\ 1 & -\alpha-\lambda \end{vmatrix} = 0$$

$$(\alpha-\lambda)^2 - 1 = 0$$

$$\alpha - \lambda = \pm 1$$

$$\lambda = \pm 1 - \alpha$$

(b) Identify all the possible type and stability of the critical point $(0, 0)$ for all non-zero eigenvalues of $\boldsymbol{A}$.

For $\alpha > 1$, both of $\pm 1 - \alpha < 0$, so we have a stable node.

For $-1 < \alpha < 1$, one of $\pm 1 - \alpha$ is positive and the other is negative, so we have a saddle node.

For $\alpha < -1$, both of $\pm 1 - \alpha > 0$, so we have an unstable node.

For $\alpha = \pm 1$, $(0,0)$ is not the only critical point of the linear system.

(c) Consider the system when $\alpha = 0$.

(i) Express the system as a single 2nd order differential equation for $x(t)$.

$$\begin{cases} \frac{dx}{dt} = y \\ \frac{dy}{dt} = x \end{cases}$$

$$\frac{d}{dt}\left(\frac{dx}{dt}\right) = \frac{d}{dt}(y)$$

$$\frac{d^2x}{dt^2} = \frac{dy}{dt}$$

$$\frac{d^2x}{dt^2} = x$$

$$\frac{d^2x}{dt^2} - x = 0$$

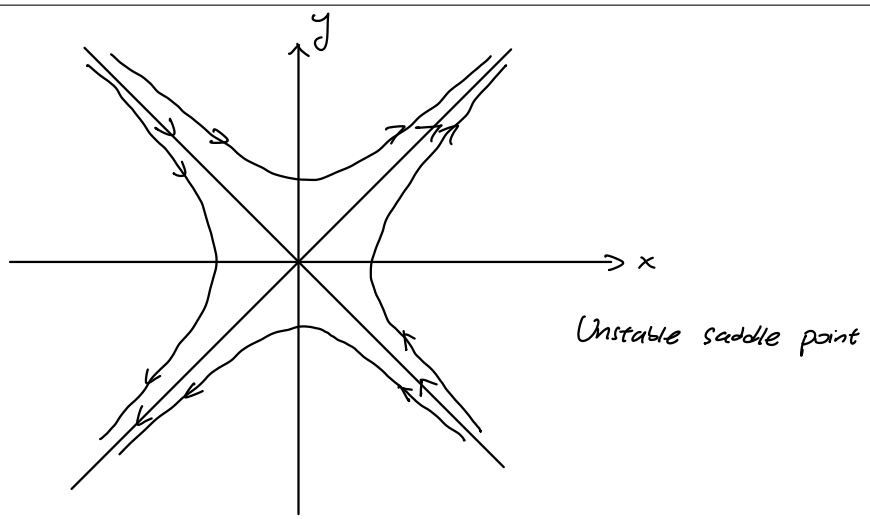
(ii) Given the system has the solution

$$\begin{bmatrix} x \\ y \end{bmatrix} = C_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + C_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-t},$$

sketch its phase portrait.

Include the following information and all relevant calculations and justification:

- any straight line orbits and their directions;
- at least 4 other orbits and their directions, showing the asymptotic behaviour as $t \to \infty$ and $t \to -\infty$;
- the slopes at which the orbits cross the $x-$ and $y-$axes;
- the loci of all minima and maxima in x, and also in y, if any of these exist;
- the orbit (or orbits) where the ratio y/x remains constant for all t and the value of this ratio;
- the value of $y(0)$, given $x(0) = a > 0$ and $\lim_{t\to\infty}(x, y) = (0, 0)$;
- the type and stability of the critical point at $(0, 0)$.



Gradient: $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{x}{y}$

$\left.\frac{dy}{dx}\right|_{x=0} = 0$

$\left.\frac{dy}{dx}\right|_{y=0}$ is undefined

$\frac{y}{x}$ is only constant along the straight line orbits $y = \pm x$, with $\frac{y}{x} = \pm 1$.

If $x(0) = a > 0$ and $\lim_{t\to\infty}(x, y) = (0, 0)$, then $y(0) = -a$ on the straight line.

$x=0$ is also the locus of extrema in y

$y=0$ is also the locus of extrema in x

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Question 2 (8 marks)

Consider the IVP

$$(t^2 + At + B)\ y''(t) + 2t\ y'(t) - 2y(t) = g(t),$$
$$y(1) = y'(1) = \pi.$$

(a) Let $A = 8, \quad B = 7, \quad g = \frac{e^t}{(2-t)}$.

Find the largest interval D where the IVP is guaranteed a unique solution that is twice differentiable.

$$y''(t) + \frac{2t}{t^2+8t+7} y'(t) - \frac{2}{t^2+8t+7} y(t) = \frac{e^t}{(2-t)(t^2+8t+7)}$$

The coefficient functions are discontinuous at $t=2,-1,-7$

As the initial condition is at $t=1$, the interval of unique solution is

$t \in (-1,2)$

(b) Let $A = B = 0, \quad g = 0$.

Given $v(t) = t$ is a solution, use reduction of order to solve the IVP.

$$t^2 y''(t) + 2t y'(t) - 2y(t) = 0$$

By reduction of order, let $y(t) = t u(t)$ and solve for $u(t)$.

$y'(t) = u(t) + t u'(t)$

$y''(t) = 2u'(t) + t u''(t)$

$$2t^2 u'(t) + t^3 u''(t) + 2t u(t) + 2t^2 u'(t) - 2t u(t) = 0$$

$$t^3 u''(t) + 4t^2 u'(t) = 0$$

$$u''(t) = -\frac{4}{t} u'(t)$$

$$\log|u'(t)| = -4\log|t| + c$$

$$u'(t) = c_1 t^{-4}$$

$$u(t) = -\frac{1}{3} c_1 t^{-3} + c_2$$

$$\Rightarrow y(t) = -\frac{1}{3} c_1 t^{-2} + c_2 t$$

$$y'(t) = \frac{2}{3} c_1 t^{-3} + c_2$$

$y(1) = \pi$ $\qquad$ $y'(1) = \pi$

$-\frac{1}{3} c_1 + c_2 = \pi$ $\qquad$ $\frac{2}{3} c_1 + c_2 = \pi$

$\Rightarrow c_1 = 0, \ c_2 = \pi$

$\Rightarrow y(t) = \pi t$

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Question 3 (16 marks)

Use Laplace transforms to find $y(t)$ for each of the following.

(a)

$$y''(t) + 9y(t) = f(t)$$

(i) $f(t)$ is a unit impulse force applied at $t = 1$ and $y(0) = 0, \quad y(\pi/4) = 0$.

$f(t) = \delta(t-1)$

$\Rightarrow y''(t) + 9y(t) = \delta(t-1)$

$\mathcal{L}[y''(t) + 9y(t)] = \mathcal{L}[\delta(t-1)]$

$s^2Y(s) - sy(0) - y'(0) + 9Y(s) = e^{-s}$, $Y(s) = \mathcal{L}[y(t)]$

$(s^2+9)Y(s) = e^{-s} + y'(0)$

$Y(s) = \frac{e^{-s}}{s^2+9} + \frac{y'(0)}{s^2+9}$

$y(t) = \frac{1}{3}\sin(3(t-1))H(t-1) + \frac{1}{3}y'(0)\sin(3t)$

$y(\frac{\pi}{4}) = 0$

$\frac{1}{3}y'(0)\sin(\frac{3\pi}{4}) = 0$, notice $\frac{\pi}{4} < 1$

$y'(0) = 0$

$\Rightarrow y(t) = \frac{1}{3}\sin(3(t-1))H(t-1)$

(ii) $f(t)$ is a unit force that is switched on at $t = 1$ and $y(0) = 0, \quad y(\pi/2) = \sin(3)$.

$f(t) = H(t-1)$

$\Rightarrow y''(t) + 9y(t) = H(t-1)$

$\mathcal{L}[y''(t) + 9y(t)] = \mathcal{L}[H(t-1)]$

$s^2Y(s) - sy(0) - y'(0) + 9Y(s) = \frac{e^{-s}}{s}$

$(s^2+9)Y(s) = \frac{e^{-s}}{s} + y'(0)$

$Y(s) = \frac{e^{-s}}{s(s^2+9)} + \frac{y'(0)}{s^2+9}$

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Partial fraction: $\frac{1}{s(s^2+9)} = \frac{A}{s} + \frac{Bs+C}{s^2+9}$

$$1 = A(s^2+9) + (Bs+C)s$$

let $s=0$: $1 = 9A$

$$A = \frac{1}{9}$$

$$\Rightarrow 1 = \frac{1}{9}s^2 + 1 + Bs^2 + Cs$$

$$\Rightarrow \begin{cases} \frac{1}{9} + B = 0 \\ C = 0 \\ 1 = 1 \end{cases} \Rightarrow \begin{cases} B = -\frac{1}{9} \\ C = 0 \end{cases}$$

$$\Rightarrow \frac{1}{s(s^2+9)} = \frac{1}{9s} - \frac{s}{9(s^2+9)}$$

$$Y(s) = \frac{1}{9}e^{-s}\left(\frac{1}{s} - \frac{s}{s^2+9}\right) + \frac{y'(0)}{s^2+9}$$

$$y(t) = \frac{1}{9}(1 - \cos(3(t-1)))H(t-1) + \frac{1}{3}y'(0)\sin(3t)$$

$$y(\tfrac{\pi}{2}) = \sin(3)$$

$$\frac{1}{9}(1 - \cos(3(\tfrac{\pi}{2}-1))) + \frac{1}{3}y'(0)\sin(\tfrac{3\pi}{2}) = \sin(3)$$, notice $\frac{\pi}{2} > 1$

$$\frac{1}{9} - \frac{1}{9}(\cos(\tfrac{3\pi}{2})\cos(3) + \sin(\tfrac{3\pi}{2})\sin(3)) - \frac{1}{3}y'(0) = \sin(3)$$

$$\frac{1}{9} + \frac{1}{9}\sin(3) - \frac{1}{3}y'(0) = \sin(3)$$

$$1 + \sin(3) - 3y'(0) = 9\sin(3)$$

$$-3y'(0) = 8\sin(3) - 1$$

$$y'(0) = \frac{1}{3}(1 - 8\sin(3))$$

$$\Rightarrow y(t) = \frac{1}{9}(1 - \cos(3(t-1)))H(t-1) + \frac{1}{9}(1 - 8\sin(3))\sin(3t)$$

(b)

$$y''(t) \;+\; \int_0^t y'(\tau)e^{2(t-\tau)}d\tau \;=\; e^{2t}$$

$$y(0) = y'(0) = 0.$$

$$\mathcal{L}[y''(t) + \int_0^t y'(\tau)e^{2(t-\tau)}d\tau] = \mathcal{L}[e^{2t}]$$

$$\mathcal{L}[y''(t) + y'(t) * e^{2t}] = \mathcal{L}[e^{2t}]$$

$$s^2Y(s) - sy(0) - y'(0) + (sY(s) - y(0)) \cdot \frac{1}{s-2} = \frac{1}{s-2}$$

$$(s^2 + \frac{s}{s-2})Y(s) = \frac{1}{s-2}$$

$$\frac{s^2(s-2)+s}{s-2}Y(s) = \frac{1}{s-2}$$

$$Y(s) = \frac{1}{s(s^2-2s+1)}$$

$$= \frac{1}{s(s-1)^2}$$

Partial fraction: $\frac{1}{s(s-1)^2} = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{(s-1)^2}$

$$1 = A(s-1)^2 + Bs(s-1) + Cs$$

Let $s=1$. $1=C$ Let $s=0$. $1=A$

$$\Rightarrow 1 = s^2 - 2s + 1 + Bs^2 - Bs + s$$

$$\Rightarrow \begin{cases} 1+B=0 \\ -2-B+1=0 \\ 1=1 \end{cases} \Rightarrow B=-1$$

$$\Rightarrow \frac{1}{s(s-1)^2} = \frac{1}{s} - \frac{1}{s-1} + \frac{1}{(s-1)^2}$$

$$Y(s) = \frac{1}{s} - \frac{1}{s-1} + \frac{1}{(s-1)^2}$$

$$y(t) = 1 - e^t + te^t$$

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Question 4 (12 marks)

Use power series to solve for $y(x)$ around $x = 0$, given

$$(x^2 + 1)y'' - 4xy' + 6y = 0.$$

Trial the power series solution $y = \sum_{n=0}^{\infty} a_n x^n$, $y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$, $y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$

$$(x^2+1)\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - 4x\sum_{n=1}^{\infty} n a_n x^{n-1} + 6\sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)a_n x^n + \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=1}^{\infty} 4n a_n x^n + \sum_{n=0}^{\infty} 6 a_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)a_n x^n + \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n - \sum_{n=1}^{\infty} 4n a_n x^n + \sum_{n=0}^{\infty} 6 a_n x^n = 0$$

$$2a_2 + 6a_3 x - 4a_1 x + 6a_0 + 6a_1 x + \sum_{n=2}^{\infty} \left(n(n-1)a_n + (n+2)(n+1)a_{n+2} - 4n a_n + 6a_n\right) x^n = 0$$

$$2a_2 + 6a_0 + (2a_1 + 6a_3)x + \sum_{n=2}^{\infty} \left((n+2)(n+1)a_{n+2} + (n^2 - 5n + 6)a_n\right) x^n = 0$$

Equating coefficients:

$$\begin{cases} 2a_2 + 6a_0 = 0 \\ 2a_1 + 6a_3 = 0 \\ (n+2)(n+1)a_{n+2} + (n^2 - 5n + 6)a_n = 0 \end{cases} \Rightarrow \begin{cases} a_2 = -3a_0 \\ a_3 = -\frac{1}{3}a_1 \\ a_{n+2} = -\frac{n^2-5n+6}{(n+2)(n+1)}a_n = -\frac{(n-2)(n-3)}{(n+2)(n+1)}a_n \end{cases}$$

Thus $a_4 = 0 \cdot a_2 = 0$ $\quad a_5 = 0 \cdot a_3 = 0$

$a_6 = 0$ $\quad a_7 = 0$

$\vdots$ $\quad \vdots$

Therefore $y = a_0 + a_1 x + a_2 x^2 + a_3 x^3$

$= a_0(1 - 3x^2) + a_1(x - \frac{1}{3}x^3)$

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Question 5 (30 marks)

A rod of unit length is initially heated to a temperature of $f(x)$. The rod is insulated except at its ends. The temperature distribution along the rod $\phi(x,t)$, for a given position $x \in [0,1]$ and time $t \in (0,\infty)$, is governed by

$$\frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial x^2}$$

subject to

$$\phi(0,t) = 0, \qquad \frac{\partial \phi}{\partial x}(1,t) = 1, \qquad t > 0$$

$$\phi(x,0) = f(x) = x + 5\sin\left(\frac{5\pi x}{2}\right), \quad 0 \le x \le 1.$$

Assume

$$\phi(x,t) = \phi_s(x) + v(x,t).$$

(a) Find the steady state temperature ϕ_s.

As $\phi(x,t) = \phi_s(x) + v(x,t)$, and the equation $\frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial x^2}$ is linear.

Then both $\phi_s(x)$ and $v(x,t)$ satisfy the heat equation.

let $\phi_s(x)$ to carry the inhomogeneous boundary condition, so we have.

$\phi_s''(x) = 0 \qquad \phi_s(0) = 0 \qquad \phi_s'(1) = 1$

$\Rightarrow \phi_s(x) = C_1 x + C_2$

$\phi_s(0) = 0 \qquad \phi_s'(1) = 1$

$C_2 = 0 \qquad C_1 = 1$

$\Rightarrow \phi_s(x) = x$

(b) Write down the boundary conditions for the transient temperature v.

$\phi(0,t) = 0$

$\phi_s(0) + v(0,t) = 0$

$0 + v(0,t) = 0$

$v(0,t) = 0$

$\phi_x(1,t) = 1$

$\phi_s'(1) + v_x(1,t) = 1$

$1 + v_x(1,t) = 1$

$v_x(1,t) = 0$

(c) Use the method of Separation of Variables to find the general solution for v.

The initial condition of $v(x,t)$ is given by:

$\phi(x,0) = x + 5\sin(\frac{5\pi x}{2})$

$\phi_s(x) + v(x,0) = x + 5\sin(\frac{5\pi x}{2})$

$x + v(x,0) = x + 5\sin(\frac{5\pi x}{2})$

$v(x,0) = 5\sin(\frac{5\pi x}{2})$

Thus, the problem for the transient part is:

$\frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2}$, $v(0,t) = 0$, $v_x(1,t) = 0$, $v(x,0) = 5\sin(\frac{5\pi x}{2})$

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Let $v(x,t) = X(x)T(t)$ and substitute into the equation.

$$T'X = X''T$$

$$\frac{T'}{T} = \frac{X''}{X}$$

As LHS depends on t only while RHS depends on x only,

then both sides have to be a constant, λ.

$$\frac{T'}{T} = \frac{X''}{X} = \lambda$$

The homogeneous boundary conditions become:

$v(0,t) = 0$ $\qquad$ $v_x(1,t) = 0$

$X(0)\,T(t) = 0$ $\qquad$ $X'(1)\,T(t) = 0$

$X(0) = 0$ $\qquad$ $X'(1) = 0$ as $T(t) = 0$ for all t is trivial

We solve the eigenvalue problem of $X(x)$ first.

$$X'' - \lambda X = 0$$

For $\lambda > 0$, let $\lambda = k^2$, $k > 0$.

$$X'' - k^2 X = 0$$

$$X = A\cosh(kx) + B\sinh(kx)$$

$$X' = kA\sinh(kx) + kB\cosh(kx)$$

$X(0) = 0$ $\qquad$ $X'(1) = 0$

$A = 0$ $\qquad$ $kB\cosh(k) = 0$

$B = 0$, as $k > 0$

$\Rightarrow$ trivial case

For $\lambda = 0$

$$X'' = 0$$

$$X = Ax + B$$

$$X' = A$$

$X(0) = 0$ $\qquad$ $X'(1) = 0$

$B = 0$ $\qquad$ $A = 0$

$\Rightarrow$ trivial case

For $\lambda < 0$, let $\lambda = -k^2$, $k > 0$

$$X'' + k^2 X = 0$$

$$X = A\cos(kx) + B\sin(kx)$$

$$X' = -kA\sin(kx) + kB\cos(kx)$$

$X(0) = 0$ $\qquad$ $X'(1) = 0$

$A = 0$ $\qquad$ $kB\cos(k) = 0$

$k = \frac{\pi}{2}(2n-1)$, $n \in \mathbb{Z}^+$ for non-trivial solution

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Hence $X_n(x) = B_n \sin(\frac{\pi}{2}(2n-1)x)$, $\lambda_n = -(\frac{\pi}{2}(2n-1))^2$, $n \in \mathbb{Z}^+$

Now solve the equation of $T_n(t)$, which depends on λ_n and thus n.

$$T_n'(t) = \lambda_n T_n(t)$$

$$T_n'(t) = -(\frac{\pi}{2}(2n-1))^2 T_n(t)$$

$$T_n(t) = C_n e^{-(\frac{\pi}{2}(2n-1))^2 t} \quad , n \in \mathbb{Z}^+$$

Thus, $v_n(x,t) = B_n C_n e^{-(\frac{\pi}{2}(2n-1))^2 t} \sin(\frac{\pi}{2}(2n-1)x)$

$$= \alpha_n e^{-(\frac{\pi}{2}(2n-1))^2 t} \sin(\frac{\pi}{2}(2n-1)x) \quad , \alpha_n = B_n C_n \quad , n \in \mathbb{Z}^+$$

The general solution is:

$$v(x,t) = \sum_{n=1}^{\infty} \alpha_n e^{-(\frac{\pi}{2}(2n-1))^2 t} \sin(\frac{\pi}{2}(2n-1)x)$$

Apply the initial condition:

$$v(x,0) = 5\sin(\frac{5\pi x}{2})$$

$$\sum_{n=1}^{\infty} \alpha_n \sin(\frac{\pi}{2}(2n-1)x) = 5\sin(\frac{5\pi x}{2})$$

By orthogonality of trigonometric functions,

$\alpha_3 = 5$, and $\alpha_n = 0$ for $n \in \mathbb{Z}^+ \setminus \{3\}$

Therefore $v(x,t) = 5e^{-(\frac{5\pi}{2})^2 t} \sin(\frac{5\pi x}{2})$

$$= 5e^{-\frac{25\pi^2 t}{4}} \sin(\frac{5\pi x}{2})$$

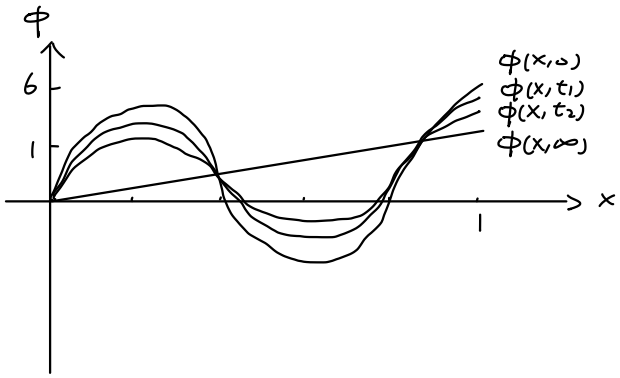
(d) Find the solution for ϕ.

$$\phi(x,t) = \phi_s(x) + v(x,t)$$
$$= x + 5e^{-\frac{25\pi^2 t}{4}} \sin\left(\frac{5\pi x}{2}\right)$$

(e) Sketch $\phi(x,0)$, $\phi(x,t_1)$, $\phi(x,t_2)$ and $\phi(x,\infty)$, $0 < t_1 < t_2 < \infty$ on one set of axes.

Period of $\sin\left(\frac{5\pi x}{2}\right)$: $\frac{2\pi}{\frac{5\pi}{2}} = 2\pi \cdot \frac{2}{5\pi} = \frac{4}{5}$

$\Rightarrow \frac{5}{4}$ cycles over $x \in [0,1]$



(f) Consider $\lim_{t\to\infty} \phi(x,t)$.

(i) Find the temperature at the midpoint of the rod.

$$\lim_{t\to\infty} \phi(x,t) = x$$

$$\lim_{t\to\infty} \phi(\tfrac{1}{2},t) = \tfrac{1}{2}$$

(ii) Find the total heat in the rod. Not in 2025

$$\int_0^1 \phi(x,t)\,dx = \int_0^1 x + 5e^{-\frac{25\pi^2 t}{4}} \sin\left(\frac{5\pi x}{2}\right) dx$$

$$= \left[\frac{1}{2}x^2 - \frac{2}{\pi} e^{-\frac{25\pi^2 t}{4}} \cos\left(\frac{5\pi x}{2}\right)\right]_0^1$$

$$= \frac{1}{2} + \frac{2}{\pi} e^{-\frac{25\pi^2 t}{4}}$$

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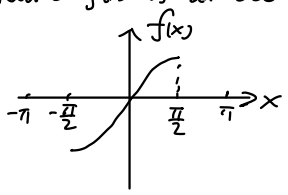
Question 6 (18 marks)

(a) Consider

$$f(x) = \begin{cases} 0 & -\pi < x < -\frac{\pi}{2} \\ \sin x & -\frac{\pi}{2} \le x \le \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < x \le \pi. \end{cases}$$

Find the Fourier series of $f(x),\ x \in (-\pi, \pi]$.

Notice $f(x)$ is an odd function:



Hence $f(x)$ has a Fourier sine series.

$$S_f(x) = \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin(nx)\,dx$$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \sin(x)\sin(nx)\,dx$$

$$= \frac{1}{\pi}\int_0^{\frac{\pi}{2}} \cos((n-1)x) - \cos((n+1)x)\,dx$$

$$= \frac{1}{\pi}\left[\frac{1}{n-1}\sin((n-1)x) - \frac{1}{n+1}\sin((n+1)x)\right]_0^{\frac{\pi}{2}}, \text{ for } n \neq 1$$

$$= \frac{1}{\pi}\left(\frac{1}{n-1}\sin\left(\frac{(n-1)\pi}{2}\right) - \frac{1}{n+1}\sin\left(\frac{(n+1)\pi}{2}\right)\right)$$

$$= \frac{1}{\pi}\left(\frac{1}{n-1}\left(\sin\left(\frac{n\pi}{2}\right)\cos\left(\frac{\pi}{2}\right) - \cos\left(\frac{n\pi}{2}\right)\sin\left(\frac{\pi}{2}\right)\right) - \frac{1}{n+1}\left(\sin\left(\frac{n\pi}{2}\right)\cos\left(\frac{\pi}{2}\right) + \cos\left(\frac{n\pi}{2}\right)\sin\left(\frac{\pi}{2}\right)\right)\right)$$

$$= -\frac{1}{\pi}\left(\frac{1}{n-1} + \frac{1}{n+1}\right)\cos\left(\frac{n\pi}{2}\right)$$

$$= -\frac{n+1+n-1}{\pi(n^2-1)}\cos\left(\frac{n\pi}{2}\right)$$

$$= -\frac{2n}{\pi(n^2-1)}\cos\left(\frac{n\pi}{2}\right)$$

$$\Rightarrow S_f(x) = \sum_{n=1}^{\infty} -\frac{2n}{\pi(n^2-1)}\cos\left(\frac{n\pi}{2}\right)\sin(nx)$$

$$= \sum_{n=1}^{\infty} -\frac{4n(-1)^n}{\pi^2(4n^2-1)}\sin(2nx)$$

(b) The linear operator

$$\mathsf{T} = -\frac{d^2}{dx^2} + 2\frac{d}{dx}$$

with the boundary conditions

$$y'(0) = y'(1) = 0$$

has eigenfunctions

$$y_n(x) = \mathrm{e}^x \cos n\pi x, \quad n = 1, 2, 3, ...$$

Find the eigenvalue λ_n for each eigenfunction y_n.

$$Ty_n(x) = -\frac{d^2}{dx^2}(e^x\cos(n\pi x)) + 2\frac{d}{dx}(e^x\cos(n\pi x))$$

$$= -\frac{d}{dx}(e^x\cos(n\pi x) - n\pi e^x\sin(n\pi x)) + 2e^x\cos(n\pi x) - 2n\pi e^x\sin(n\pi x)$$

$$= -e^x\cos(n\pi x) + n\pi e^x\sin(n\pi x) + n\pi e^x\sin(n\pi x) + n^2\pi^2 e^x\cos(n\pi x)$$
$$+ 2e^x\cos(n\pi x) - 2n\pi e^x\sin(n\pi x)$$

$$= e^x\cos(n\pi x) + n^2\pi^2 e^x\cos(n\pi x)$$

$$= (1+n^2\pi^2)\, y_n(x)$$

Hence $\lambda_n = 1 + n^2\pi^2$

(c) The Fourier series of $x^4, x \in (-\pi, \pi]$ is given by

$$s(x) = \frac{\pi^4}{5} + \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \left(8\pi^2 - \frac{48}{n^2}\right) \cos nx.$$

Use this result to find

$$\sum_{n=1}^{\infty} \left(\frac{1}{n^4} - \frac{\pi^2}{6}\frac{1}{n^2}\right).$$

$$s(\pi) = \frac{\pi^4}{5} + \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\left(8\pi^2 - \frac{48}{n^2}\right)(-1)^n$$

$$= \frac{\pi^4}{5} + \sum_{n=1}^{\infty}\left(\frac{8\pi^2}{n^2} - \frac{48}{n^4}\right)$$

The even extension of x^4 is continuous at $x = \pi$.

Hence $s(\pi) = \pi^4 = \frac{\pi^4}{5} + \sum_{n=1}^{\infty}\left(\frac{8\pi^2}{n^2} - \frac{48}{n^4}\right)$ from convergence of Fourier series

$$\frac{4\pi^4}{5} = \sum_{n=1}^{\infty}\left(\frac{8\pi^2}{n^2} - \frac{48}{n^4}\right)$$

$$-\frac{\pi^4}{60} = \sum_{n=1}^{\infty}\left(\frac{1}{n^4} - \frac{\pi^2}{6n^2}\right)$$

$$\Rightarrow \sum_{n=1}^{\infty}\left(\frac{1}{n^4} - \frac{\pi^2}{6n^2}\right) = -\frac{\pi^4}{60}$$

(d) Consider

$$y''(x) - 2y(x) = s(x)$$

subject to periodic boundary conditions at $x = \pm\pi$.

Given $s(x)$ is the Fourier series of $x^2,\ x \in (-\pi, \pi]$,

$$s(x) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx,$$

find a *particular solution* to the ODE by trialling an arbitrary Fourier series. **You are not required to find the complementary solution.**

Trial a Fourier series $y(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos(nx) + b_n \sin(nx)$.

$$y'(x) = \sum_{n=1}^{\infty}(-na_n \sin(nx) + nb_n \cos(nx))$$

$$y''(x) = \sum_{n=1}^{\infty}(-n^2 a_n \cos(nx) - n^2 b_n \sin(nx))$$

$$-a_0 + \sum_{n=1}^{\infty}(-n^2 a_n \cos(nx) - n^2 b_n \sin(nx) - 2a_n \cos(nx) - 2b_n \sin(nx)) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(nx)$$

$$\Rightarrow \begin{cases} -a_0 = \frac{\pi^2}{3} \\ -(n^2+2)a_n = \frac{4(-1)^n}{n^2} \\ -(n^2+2)b_n = 0 \end{cases} \Rightarrow \begin{cases} a_0 = -\frac{\pi^2}{3} \\ a_n = -\frac{4(-1)^n}{n^2(n^2+2)} \\ b_n = 0 \end{cases}$$

$$\Rightarrow y(x) = -\frac{\pi^2}{6} - \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2(n^2+2)} \cos(nx)$$

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Question 7 (12 marks)

(a) Use the integral definition of the Fourier transform to evaluate

$$\mathscr{F}\Big(\delta(x-a)\Big), \qquad a \in \mathbb{R}.$$

$$\mathcal{F}[\delta(x-a)] = \int_{-\infty}^{\infty} \delta(x-a) e^{-ikx}\, dx$$
$$= e^{-ika}$$

(b) Consider the 1D advection PDE for $u(x,t)$,

$$\frac{\partial u}{\partial t} = c\frac{\partial u}{\partial x}, \qquad x \in \mathbb{R},\ t > 0,\ c \in \mathbb{R},$$

subject to the initial condition $u(x,0) = \delta(x)$.

Use Fourier transforms to find u.

Take Fourier transform of the equation and the initial condition with respect to x.

Denote $\hat{u}(k,t) = \mathcal{F}[u(x,t)]$

$$\mathcal{F}\left[\frac{\partial u}{\partial t}\right] = \mathcal{F}\left[c\frac{\partial u}{\partial x}\right] \qquad \mathcal{F}[u(x,0)] = \mathcal{F}[\delta(x)]$$

$$\frac{\partial \hat{u}}{\partial t} = -ick\hat{u} \qquad \hat{u}(k,0) = 1$$

$$\hat{u} = C(k) e^{-ickt}$$

$$\hat{u}(k,0) = C(k) = 1$$

$$\Rightarrow \hat{u}(k,t) = e^{-ickt}$$

$$u(x,t) = \mathcal{F}^{-1}[e^{-ickt}]$$
$$= \delta(x-ct) \quad , \text{as } \mathcal{F}^{-1}[e^{-ika}] = \delta(x-a)$$

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(c) Use the following Fourier transform

$$\mathscr{F}\left(\mathrm{e}^{-ax^2}\right) = \sqrt{\frac{\pi}{a}}\ \mathrm{e}^{-\frac{k^2}{4a}}, \qquad a > 0,$$

to evaluate

$$\int_{-\infty}^{\infty} \cos(2x)\ \mathrm{e}^{-x^2/2} dx.$$

$$\int_{-\infty}^{\infty} \cos(2x) e^{-\frac{1}{2}x^2}\, dx = \frac{1}{2}\left(\int_{-\infty}^{\infty} e^{2ix} e^{-\frac{1}{2}x^2}\, dx + \int_{-\infty}^{\infty} e^{-2ix} e^{-\frac{1}{2}x^2}\, dx\right)$$

$$= \frac{1}{2}\left(\mathcal{F}\left[e^{-\frac{1}{2}x^2}\right]\Big|_{k=-2} + \mathcal{F}\left[e^{-\frac{1}{2}x^2}\right]\Big|_{k=2}\right)$$

$$= \frac{1}{2}\left(\sqrt{2\pi}\, e^{-\frac{1}{2}k^2}\Big|_{k=-2} + \sqrt{2\pi}\, e^{-\frac{1}{2}k^2}\Big|_{k=2}\right)$$

$$= \frac{\sqrt{2\pi}}{2}\left(e^{-2} + e^{-2}\right)$$

$$= \sqrt{2\pi}\, e^{-2}$$

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End of Exam — Total Available Marks = 120

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MAST20030 Differential Equations Formulae Sheet

1) **Laplace Transforms**

1.	$f(t)$	$F(s) = \mathscr{L}(f) = \int_0^\infty f(t)e^{-st}\,dt$	(Definition of Transform)
2.	1	$\frac{1}{s}$	
3.	t^n	$\frac{n!}{s^{n+1}}$	
4.	e^{at}	$\frac{1}{s-a}$	
5.	$\sin(at)$	$\frac{a}{s^2+a^2}$	
6.	$\cos(at)$	$\frac{s}{s^2+a^2}$	
7.	$\sinh(at)$	$\frac{a}{s^2-a^2}$	
8.	$\cosh(at)$	$\frac{s}{s^2-a^2}$	
9.	$\delta(t-a)$	e^{-as}	$(a \geq 0)$
10.	$f'(t)$	$sF(s) - f(0)$	
11.	$f''(t)$	$s^2F(s) - sf(0) - f'(0)$	
12.	$f^{(n)}(t)$	$s^nF(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0)$	
13.	$\int_0^t f(\tau)\,d\tau$	$\frac{F(s)}{s}$	
14.	$e^{-at}f(t)$	$F(s+a)$	(s-Shifting Theorem)
15.	$f(t-a)\,u(t-a)$	$e^{-as}F(s)$	($a > 0$, t-Shifting Theorem) $u(t)$ is the Heaviside step function
16.	$\int_0^t f(\tau)\,g(t-\tau)\,d\tau$	$F(s)\,G(s)$	(Convolution)

2) **Standard Limits**

(i) $\lim_{n\to\infty} \frac{1}{n^p} = 0 \quad (p > 0)$

(ii) $\lim_{n\to\infty} r^n = 0 \quad (|r| < 1)$

(iii) $\lim_{n\to\infty} a^{1/n} = 1 \quad (a > 0)$

(iv) $\lim_{n\to\infty} n^{1/n} = 1$

(v) $\lim_{n\to\infty} \frac{a^n}{n!} = 0 \quad (\text{all } a)$

(vi) $\lim_{n\to\infty} \frac{\log_e n}{n^p} = 0 \quad (p > 0)$

(vii) $\lim_{n\to\infty} \left(1 + \frac{a}{n}\right)^n = \mathrm{e}^a \quad (\text{all } a)$

(viii) $\lim_{n\to\infty} \frac{n^p}{a^n} = 0 \quad (\text{all } p, a > 1)$

3) **The Generalised Harmonic Series (p-Series)**

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } p > 1 \\ \text{divergent} & \text{if } p \leq 1 \end{cases}$$

4) **Geometric Series**

$$\sum_{n=0}^{\infty} ar^n \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } |r| < 1 \\ \text{divergent} & \text{if } |r| \geq 1 \end{cases}$$

5) **Fourier Series Formulae**

$$\begin{aligned} f(x) &= \frac{1}{2}a_0 + \sum_{n=1}^{\infty} \left(a_n \cos k_n x + b_n \sin k_n x\right), \quad k_n = \frac{n\pi}{L} \\ a_0 &= \frac{1}{L}\int_{-L}^{L} f(x)\, dx \\ a_n &= \frac{1}{L}\int_{-L}^{L} f(x) \cos k_n x\, dx \\ b_n &= \frac{1}{L}\int_{-L}^{L} f(x) \sin k_n x\, dx \end{aligned}$$

6) **Fourier Transforms Formulae**

$$\widehat{f}(k) = \mathscr{F}(f) = \int_{-\infty}^{\infty} f(x)\mathrm{e}^{-ikx}\, dx$$

$$f(x) = \mathscr{F}^{-1}(\widehat{f}) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(k)\mathrm{e}^{ikx}\, dk$$

$$\mathscr{F}\left(\frac{\mathrm{d}^n f}{\mathrm{d}x^n}\right) = (ik)^n \widehat{f}, \qquad \mathscr{F}\left(x^n f\right) = i^n \frac{\mathrm{d}^n \widehat{f}}{\mathrm{d}k^n}$$

7) **Trigonometric and Hyperbolic Formulae**

$$\cos^2 x + \sin^2 x = 1$$
$$1 + \tan^2 x = \sec^2 x$$
$$\cot^2 x + 1 = \operatorname{cosec}^2 x$$

$$\cosh^2 x - \sinh^2 x = 1$$
$$1 - \tanh^2 x = \operatorname{sech}^2 x$$
$$\coth^2 x - 1 = \operatorname{cosech}^2 x$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$
$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$$
$$\cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$$

$$\sin x \sin y = \frac{1}{2}\left[\cos(x-y) - \cos(x+y)\right]$$
$$\cos x \cos y = \frac{1}{2}\left[\cos(x-y) + \cos(x+y)\right]$$
$$\sin x \cos y = \frac{1}{2}\left[\sin(x-y) + \sin(x+y)\right]$$

$$\sinh x \sinh y = \frac{1}{2}\left[\cosh(x+y) - \cosh(x-y)\right]$$
$$\cosh x \cosh y = \frac{1}{2}\left[\cosh(x+y) + \cosh(x-y)\right]$$
$$\sinh x \cosh y = \frac{1}{2}\left[\sinh(x+y) + \sinh(x-y)\right]$$

$$\sin x = \frac{i}{2}\left(\mathrm{e}^{-ix} - \mathrm{e}^{ix}\right)$$
$$\cos x = \frac{1}{2}\left(\mathrm{e}^{-ix} + \mathrm{e}^{ix}\right)$$

$$\sinh x = \frac{1}{2}\left(\mathrm{e}^{x} - \mathrm{e}^{-x}\right)$$
$$\cosh x = \frac{1}{2}\left(\mathrm{e}^{x} + \mathrm{e}^{-x}\right)$$

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